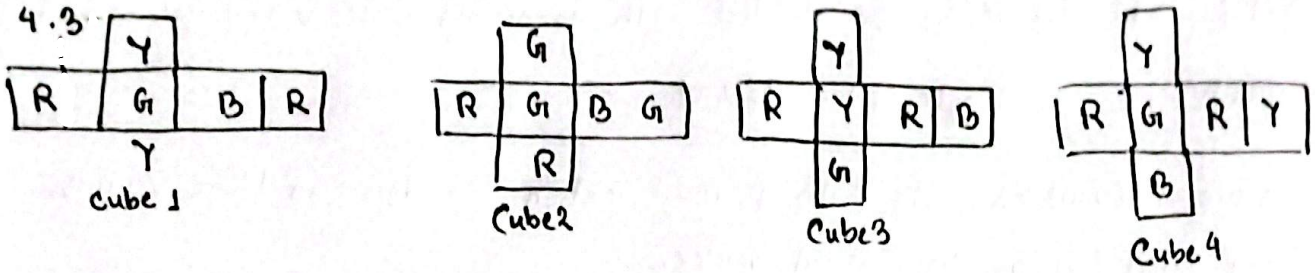
CSE 708
Graph Theory

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Id: 21301002

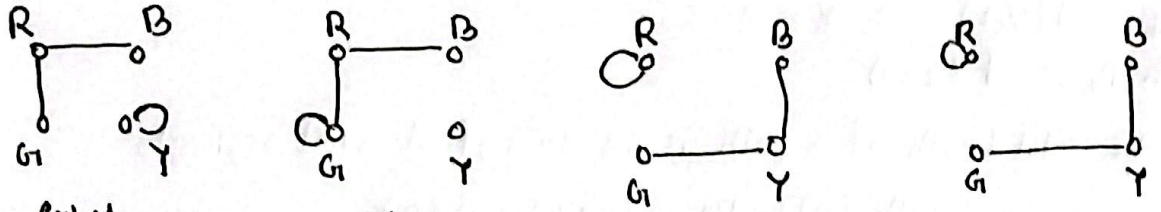
$X = \text{mod}(21301002\text{4}, 5) + 1 = 3$

4.3



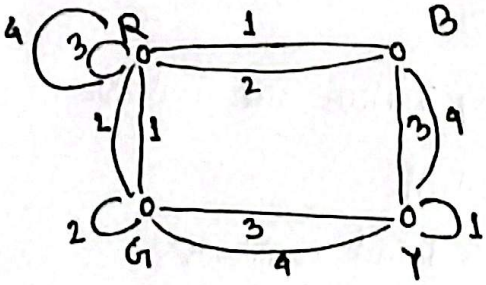
Cube 1 Cube 2 Cube 3 Cube 4

To solve the problem we will represent the graph with four vertices R, B, G, Y for each colors.

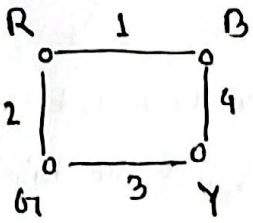


Cube1 Cube2 Cube3 Cube 4

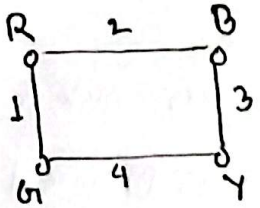
Now we will create a superimposed graph. The numbers in the superimposed graphs edges signifies the cube number. Next from this graph we will create two subgraph H_1 and H_2. H_1 tells us which color appears from front ~~two~~ and back. H_2 represent left and right.



Superimposed Graph



Front and Back (H_1)

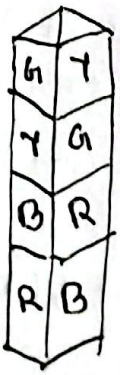


Left and Right (H_2)

From the graph H_1 and H_2 we get the correct cube stacking order



Left Front



Right Back